

MailTrace AI

Digital Email Forensic Investigation Report

1. Case Information

Case ID	MT-20260913182049-A74727
Generated At	2026-09-13 18:20:49
Filename	unseen_phishing_test.eml
Sender	Bank Security <alerts@bank-security.example>
Receiver	user@example.com
Subject	Your Account Will Be Locked Today

2. Threat Risk Assessment

Risk Score	87.69 / 100
Risk Level	CRITICAL
Primary Threat	Phishing
Recommended Action	Immediate containment, sender verification and forensic investigation recommended.

3. Machine Learning Analysis

Classification: Phishing

Confidence: 25.44%

Threat Class	Probability
Phishing	25.44%
Account Takeover	12.99%
Credential Theft	10.07%
Financial Fraud	8.64%
OTP Fraud	8.64%

4. Email Header Forensics

Authentication	Status
SPF	FAIL
DKIM	FAIL
DMARC	FAIL
Header Anomalies	5

5. Indicators of Compromise

IOC Type	Count
Emails	5
URLs	1
Domains	5

IP Addresses	0
MD5	0
SHA1	0
SHA256	0

6. Geo Intelligence

City: Hyderabad

Region: Telangana

Country: India

Latitude: 17.385

Longitude: 78.4867

Infrastructure: Demo Financial Infrastructure

7. Explainable Findings

ML Threat Classification [HIGH] — ML model identified the email as Phishing.

Header Forensics [HIGH] — Reply-To domain differs from sender domain.

Header Forensics [MEDIUM] — Return-Path domain differs from sender domain.

Header Forensics [HIGH] — SPF authentication failed.

Header Forensics [HIGH] — DKIM authentication failed.

Header Forensics [HIGH] — DMARC authentication failed.

Authentication [HIGH] — SPF authentication failed.

Authentication [HIGH] — DKIM authentication failed.

Authentication [HIGH] — DMARC authentication failed.

IOC Analysis [MEDIUM] — 11 indicators of compromise were extracted.

Content Analysis [HIGH] — Suspicious language patterns detected: Credential Phishing, Financial Fraud

Forensic Correlation [HIGH] — Multiple email authentication mechanisms failed.

Threat Correlation [HIGH] — Content analysis indicates a high-risk social-engineering or fraud-related pattern.

Identity Forensics [HIGH] — Header anomalies correlate with failed sender authentication.

8. Threat Graph Summary

Graph Nodes: 10

Graph Relationships: 9

Forensic Notice: Geo intelligence represents approximate infrastructure-level or controlled demo location data. It must not be interpreted as an exact physical location of a sender.

Generated by MailTrace AI Threat Intelligence Engine